Introduction to Computer Vision (CS641)

Lecture03 - Image Filtering

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Image noise

- Noise in image is any degradation in image signal caused by external disturbance.

$$\hat{\mathtt{I}}(x, y) = \mathtt{I}(x, y) + n(x, y)$$

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- We take average of the neighboring pixel to remove noise. Generally we take moving average.

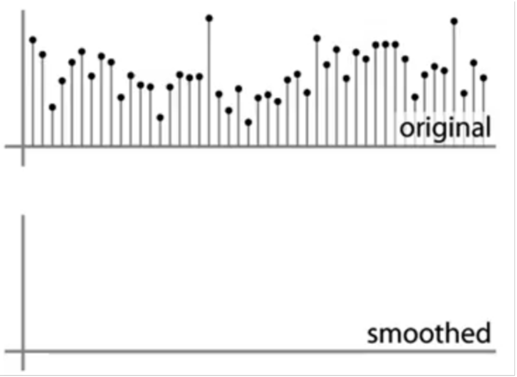
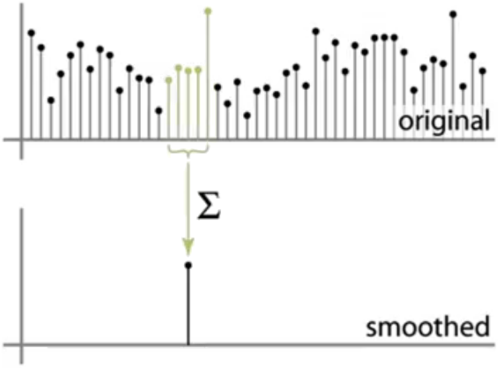


Figure: Average of 1D image. Source: S. Marschner.

- We take average of the neighboring pixel to remove noise. Generally we take moving average.



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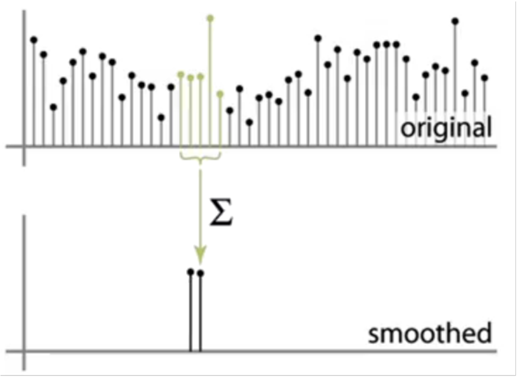

Figure: Average of 1D image. Source: S. Marschner.

Removing image noise

Averaging: ID image

- We take average of the neighboring pixel to remove noise. Generally we take moving average.

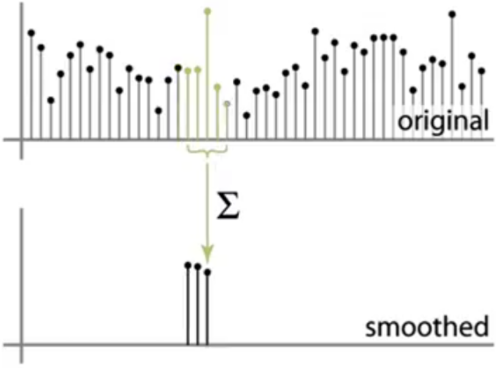

Figure: Average of 1D image. Source: S. Marschner.

Removing image noise

Averaging: ID image

- We take average of the neighboring pixel to remove noise. Generally we take moving average.

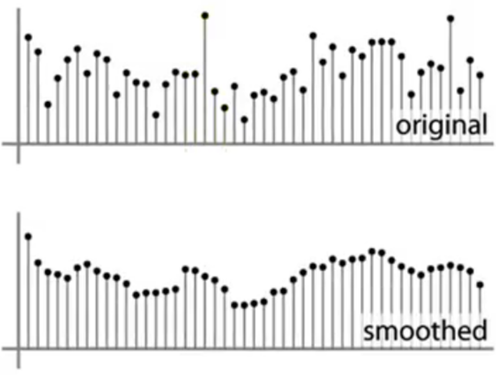

Figure: Average of 1D image. Source: S. Marschner.

- Eventually we receive the smooth version of the image.
- Why this solution is intuitive to take an average of sub region to reduce noise.

- The true value of a pixel is probably similar to its neighboring pixels.
- The noise added to each pixel is independent of the noise added to all other pixels.

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- Uniform weights [1, 1, 1, 1, 1]/5
- Just think of an idea. The more nearby the more related to me and will contribute more to the average.

ID image noise
Weighted Average

- Uniform weights [1, 1, 1, 1, 1] / 5

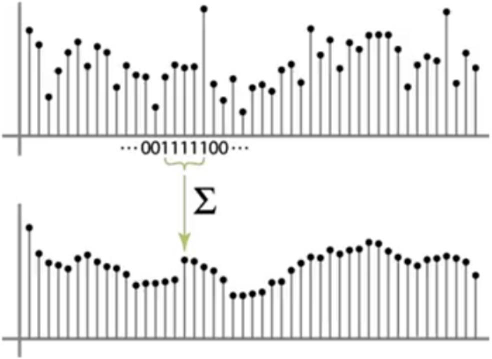

Figure: Weighted Average for 1D image noise. Source: S. Marschner.

- Non Uniform weights [1, 4, 6, 4, 1] / 16

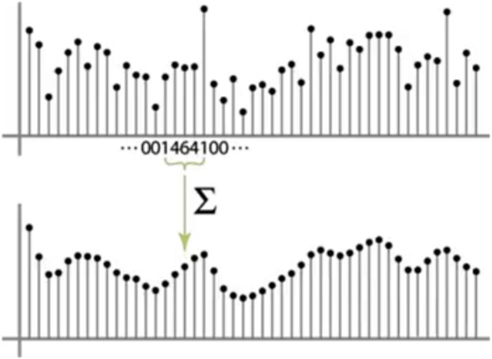

Figure: Weighted Average for 1D image noise. Source: S. Marschner.

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- Moving average in 2D using box filter.

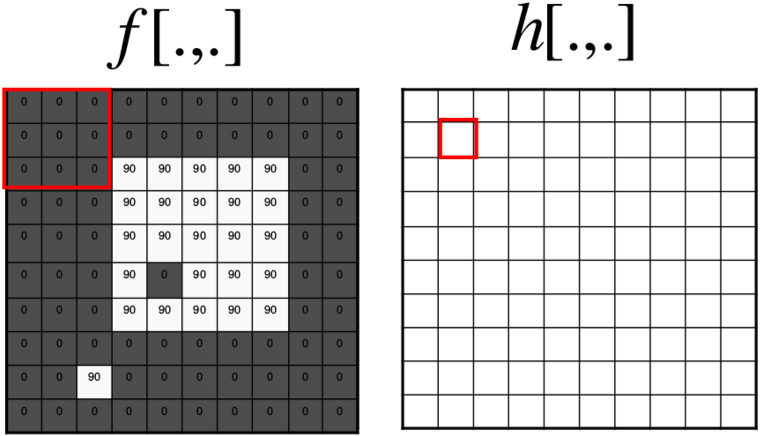

Figure: Weighted Average for 1D image noise. Source: S. Seitz.

- Moving average in 2D using box filter.

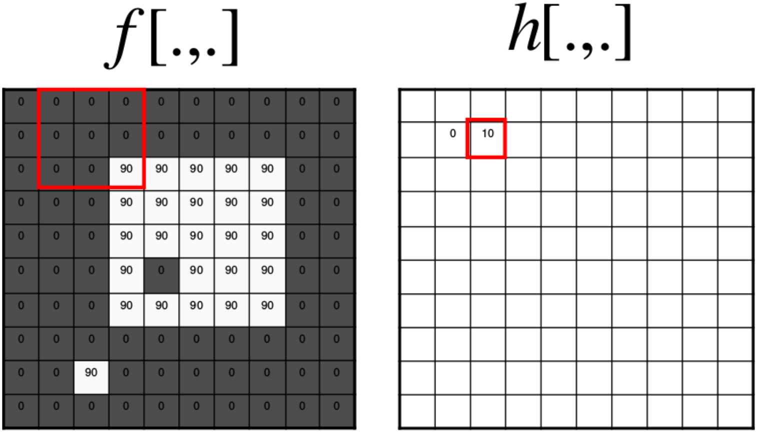

Figure: Weighted Average for 1D image noise. Source: S. Seitz.

Box filter

Moving average 2D

- Moving average in 2D using box filter.

$f[\cdot,\cdot]$

0	0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0	0
0	0	0	90	90	90	90	90	0	0
0	0	0	90	90	90	90	90	0	0
0	0	0	90	90	90	90	90	0	0
0	0	0	90	0	90	90	90	0	0
0	0	0	90	90	90	90	90	0	0
0	0	0	0	0	0	0	0	0	0
0	0	90	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0	0

$h[\cdot,\cdot]$

	0	10	20						

Figure: Weighted Average for 1D image noise. Source: S. Seitz.

Box filter
Moving average 2D

- Moving average in 2D using box filter.

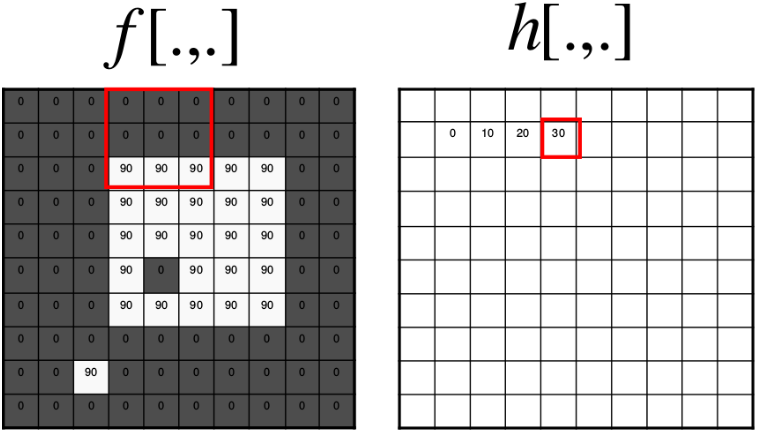

Figure: Weighted Average for 1D image noise. Source: S. Seitz.

Box filter
Moving average 2D

- Moving average in 2D using box filter.

$f[.,.]$

0	0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0	0
0	0	0	90	90	90	90	90	0	0
0	0	0	90	90	90	90	90	0	0
0	0	0	90	90	90	90	90	0	0
0	0	0	90	0	90	90	90	0	0
0	0	0	90	90	90	90	90	0	0
0	0	0	0	0	0	0	0	0	0
0	0	90	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0	0

$h[.,.]$

	0	10	20	30	30				

Figure: Weighted Average for 1D image noise. Source: S. Seitz.

Box filter

Moving average 2D

- Moving average in 2D using box filter.

$f[.,.]$

0	0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0	0
0	0	0	90	90	90	90	90	0	0
0	0	0	90	90	90	90	90	0	0
0	0	0	90	90	90	90	90	0	0
0	0	0	90	0	90	90	90	0	0
0	0	0	90	90	90	90	90	0	0
0	0	0	0	0	0	0	0	0	0
0	0	90	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0	0

$h[.,.]$

	0	10	20	30	30				
				?					

Figure: Weighted Average for 1D image noise. Source: S. Seitz.

- Moving average in 2D using box filter.

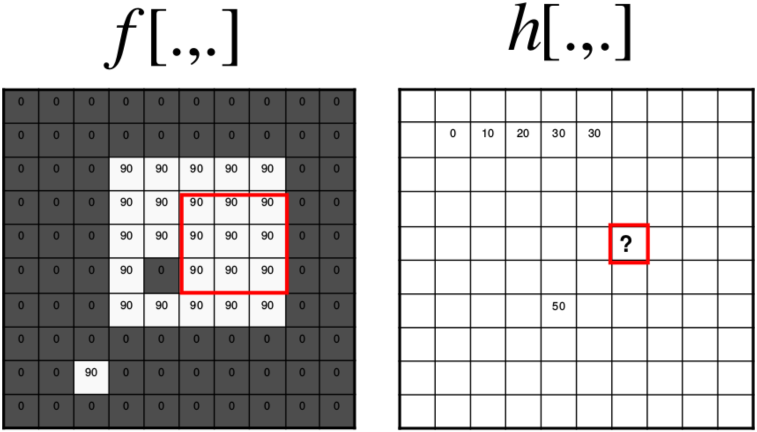

Figure: Weighted Average for 1D image noise. Source: S. Seitz.

Box filter
Moving average 2D

- Moving average in 2D using box filter.

$f[.,.]$

0	0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0	0
0	0	0	90	90	90	90	90	0	0
0	0	0	90	90	90	90	90	0	0
0	0	0	90	90	90	90	90	0	0
0	0	0	90	0	90	90	90	0	0
0	0	0	90	90	90	90	90	0	0
0	0	0	0	0	0	0	0	0	0
0	0	90	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0	0

$h[.,.]$

	0	10	20	30	30	30	20	10	
	0	20	40	60	60	60	40	20	
	0	30	60	90	90	90	60	30	
	0	30	50	80	80	90	60	30	
	0	30	50	80	80	90	60	30	
	0	20	30	50	50	60	40	20	
	10	20	30	30	30	30	20	10	
	10	10	10	0	0	0	0	0	

Figure: Weighted Average for 1D image noise. Source: S. Seitz.

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Correlation

Uniform weights

- Loop over all pixels in neighborhood around pixel $f(i,j)$, and summing up all the pixels.
- Divide the whole thing by the number of weights in the filter and uniform for each pixel.
- The averaging window size is $2k+1 \times 2k+1$. The window size will be be odd.

$$h(i,j) = \frac{1}{(2n+1)^2} \sum_{k=-n}^{n} \sum_{l=-n}^{n} f(i+k, j+l)$$

- Now generalize to allow different weights depending on neight
- This is also called cross correlation and denoted $h = gf$

$$h(i,j) = \sum_{l=-n}^{n} g(k,l) f(i+k, j+l)$$

- $g(k,l)$ is the non-uniform weights and it referred is kernel or mask.
- The filter kernel or mask $g(k,l)$ is the matrix of weights in the linear combination.

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- Replaces each pixel with an average of its neighborhood.
- Achieve smoothing effect that is to remove the sharp features.

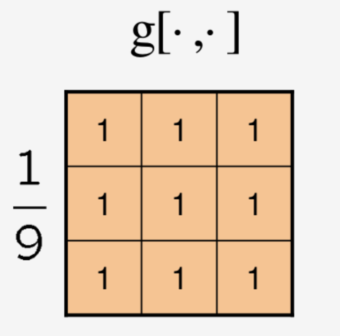
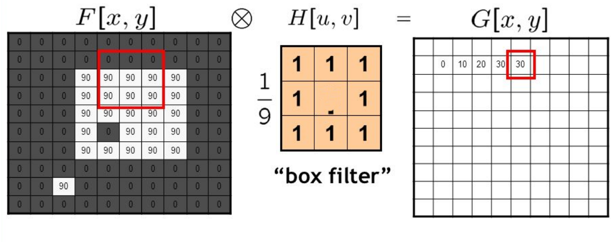


Figure: The box filter. Slide source: David Lowe (UBC).

- The values belong in the kernel $h(u, v)$ for the moving average example.



$$G = H \otimes F$$

Figure: The box filter. Source: Kristen Grauman.

Contents

Filtering

Gaussian filter

- A Gaussian kernel gives less weight to pixels further form the center of the window.
- Removes the high-frequency components from image a “low pass filter.”

0	0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0	0
0	0	0	90	90	90	90	90	0	0
0	0	0	90	90	90	90	90	0	0
0	0	0	90	90	90	90	90	0	0
0	0	0	90	0	90	90	90	0	0
0	0	0	90	90	90	90	90	0	0
0	0	0	0	0	0	0	0	0	0
0	0	90	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0	0

$F[x, y]$

$$\frac{1}{16}\begin{bmatrix} 1 & 2 & 1 \\ 2 & 4 & 2 \\ 1 & 2 & 1 \end{bmatrix}$$

$H[u, v]$

This kernel is an approximation of a 2d Gaussian function:

$$h(u, v) = \frac{1}{2\pi\sigma^2} e^{-\frac{u^2+v^2}{\sigma^2}}$$

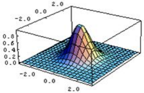
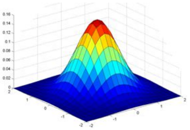


Figure: The Gaussian filter. Source: S. Seitz

Filtering
Gaussian filter

- Weight contribution of neighboring pixels by nearness.



0.003	0.013	0.022	0.013	0.003
0.013	0.059	0.097	0.059	0.013
0.022	0.097	0.159	0.097	0.022
0.013	0.059	0.097	0.059	0.013
0.003	0.013	0.022	0.013	0.003

5 x 5, σ = 1

$$G_\sigma = \frac{1}{2\pi\sigma^2} e^{-\frac{(x^2+y^2)}{2\sigma^2}}$$

Figure: The Gaussian filter. Source: S. Seitz

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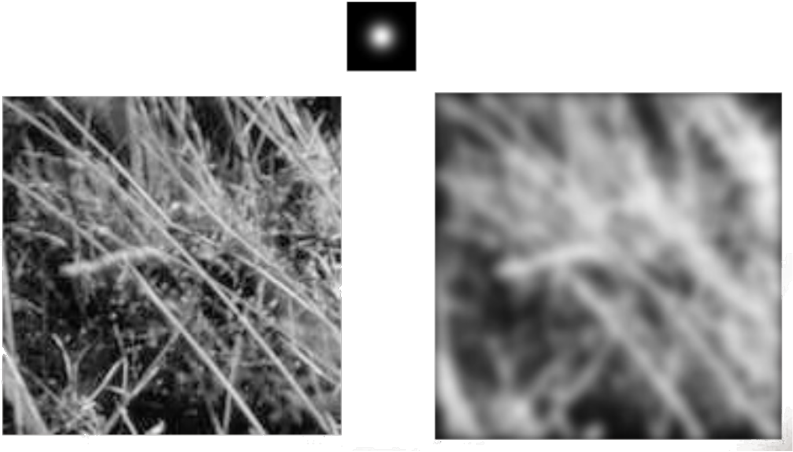


Figure: Smoothing with the Gaussian filter. Source: S. Seitz

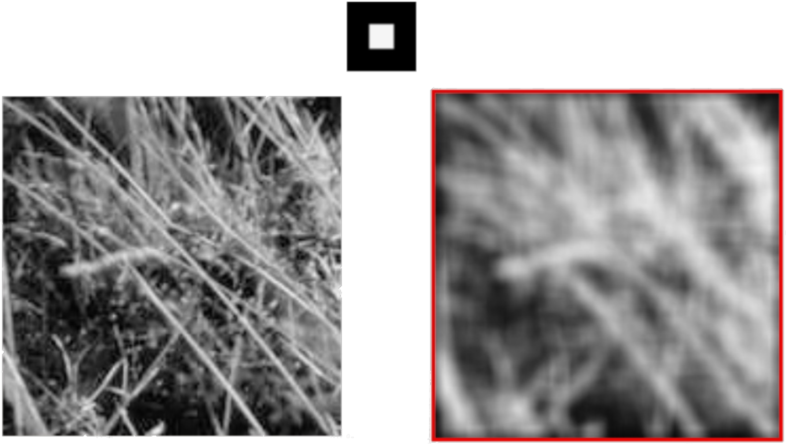
- 

Figure: Smoothing with the Gaussian filter. Source: S. Seitz

Next class - Edge Detection